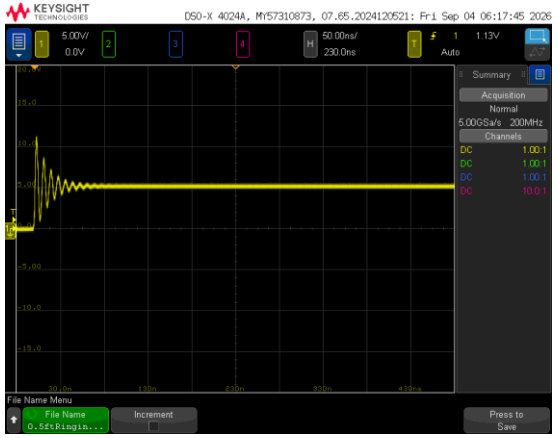
Lab 1.3

Hacking Ringing Reflections in Coax Cables

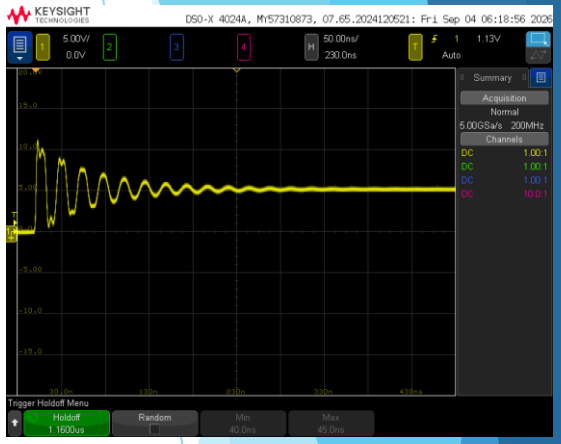
Leo Marcinkus and Jordan Bernhardt

Setup and Measurement

- Board 100, 3 Ω output, 5 V supply -> RG58 coax -> scope 1 MΩ input (no probe)
  - Two cable lengths, same cable type: 0.5 ft and 3 ft
- Scope: Keysight DSO 4024A, 5 GSa/s, 200 MHz
  - 50 ns/div for both captures so the two data sets share one time base
- **Trigger:** Rising edge at 1.13 V
  - 1.16 µs trigger holdoff for the 3 ft cable
  - 0.96 µs trigger holdoff for the 0.5 ft cable
  - Holdoff helps accomplish a lower threshold
  - Low threshold puts the start of the edge just before t = 0 (ADS only plots t ≥ 0)
- **Best practices applied:** Averaging off while setting up time base long enough to see the steady state (500 ns), 2000 points at 0.25 ns saved as CSV



0.5 ft cable, 50 ns/div

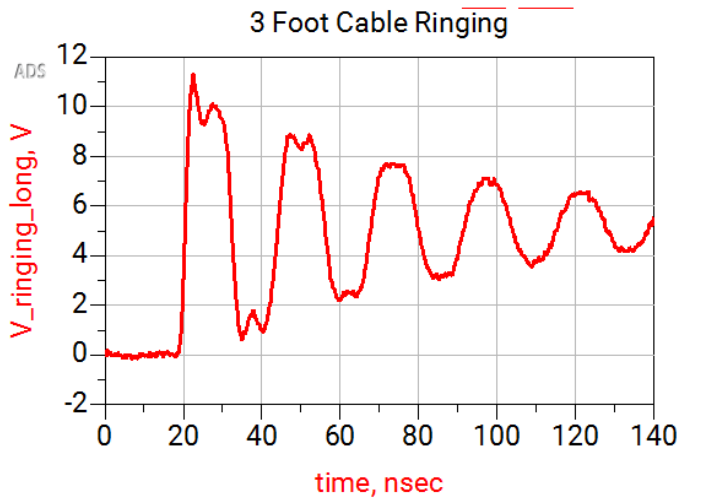


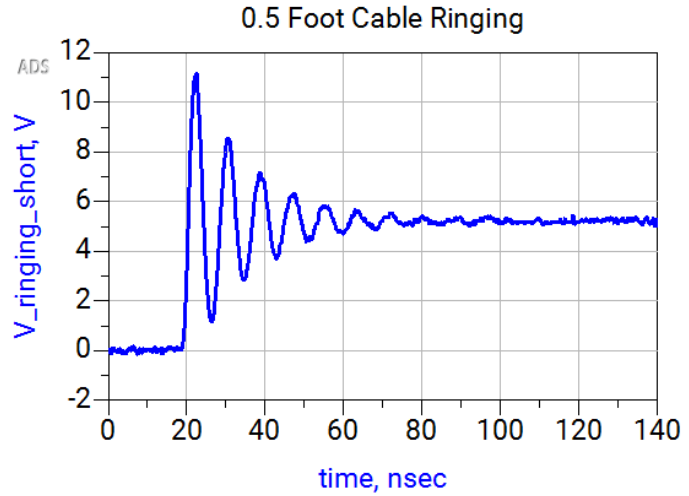
3 ft cable, 50 ns/div

What the Measurements Say

- **Same first peak, 11.1 V, and same steady state, 5.24 V**
- **Ring period:** 8.3 ns for 0.5 ft, 24.5 ns for 3 ft. Period ≈ 4 x TD
  - TD ≈ 2.1 ns and 6.1 ns
- Cable alone would give 0.77 ns and 4.6 ns (1.54 ns/ft). Both cables carry ≈ 1.3-1.5 ns extra: the SMA-BNC adapter, connectors, and the scope's input capacitance and resistance
  - 2.1 ns – 0.77 ns = 1.3 ns, 6.1 ns – 4.6 ns = 1.5 ns
- **Delay per foot from the difference:** (6.1 - 2.1) ns / (3 – 0.5) ft = 1.6 ns/ft
- **Decay:** peak-to-trough shrinks by 0.74–0.78 per period -> ρ_{src} ≈ -0.87 -> effective R_{src} ≈ 3-4 Ω, above the 1.7 Ω DC value from Lab 1.1. The lossless model must put the cable loss somewhere

Feature	0.5 ft	3 ft
1st peak / steady state	11.09 V / 5.24 V	11.11 V / 5.22 V
Ring period	8.3 ns	24.5 ns
Effective TD = period / 4	2.1 ns	6.1 ns
Cable-only TD (RG58 Spec: 1.54 ns/ft)	0.77 ns	4.6 ns
Extra delay (fixture + scope C)	1.3 ns	1.5 ns
R_{src} from decay	3.7 Ω	3.0 Ω





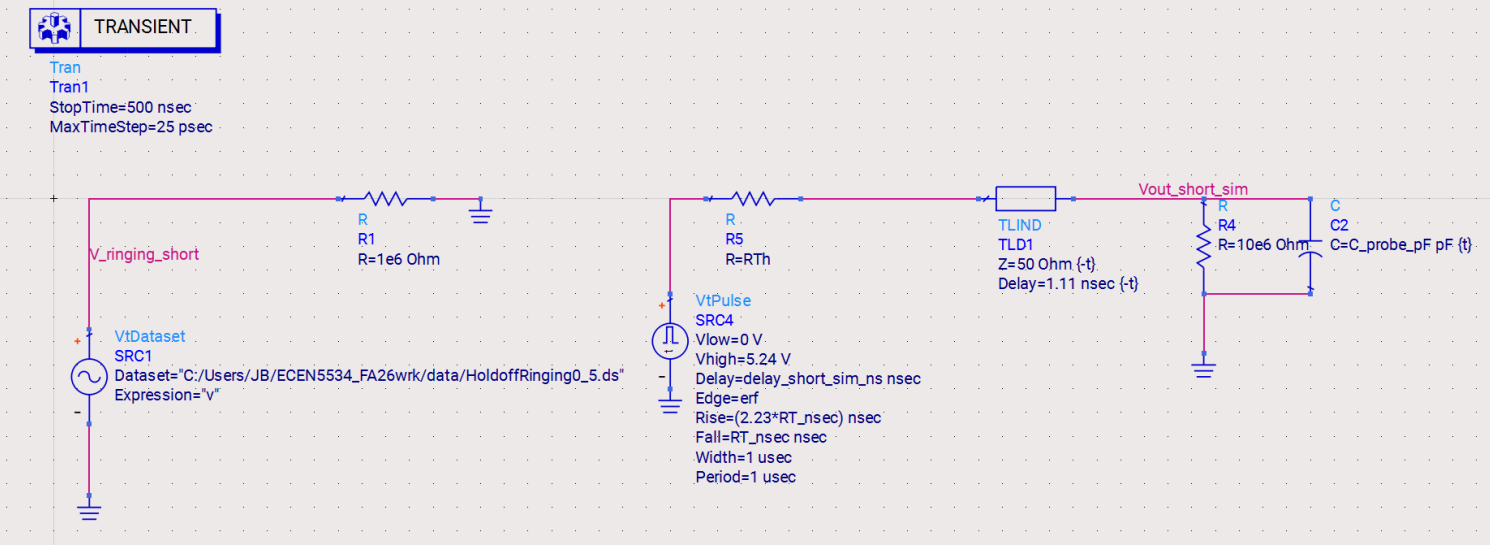
A subset of the measured data as imported into ADS the same data as collected from the scope

Bringing the Waveforms into ADS

- **Step 1 - header:** Replace the two scope header lines with BEGIN BLK and % time(1) v(1)
  - Keep time ≥ 0 (we shifted by +20 ns)
- **Step 2 - Dataset:** Tools -> Data File Utility -> Data File Tool, read as MDIF / Generic MDIF, write .ds
- **Step 3 - Source:** VtDataset with Expression = "v", output tied to ground through a dummy resistor so no node floats
- **Step 4 - Transient:** Start 0, Stop 500 ns, Min = Max step = 0.25 ns = the scope sample intervals
- **Rule #9:** The imported waveform must look exactly like the scope screen before any model is added

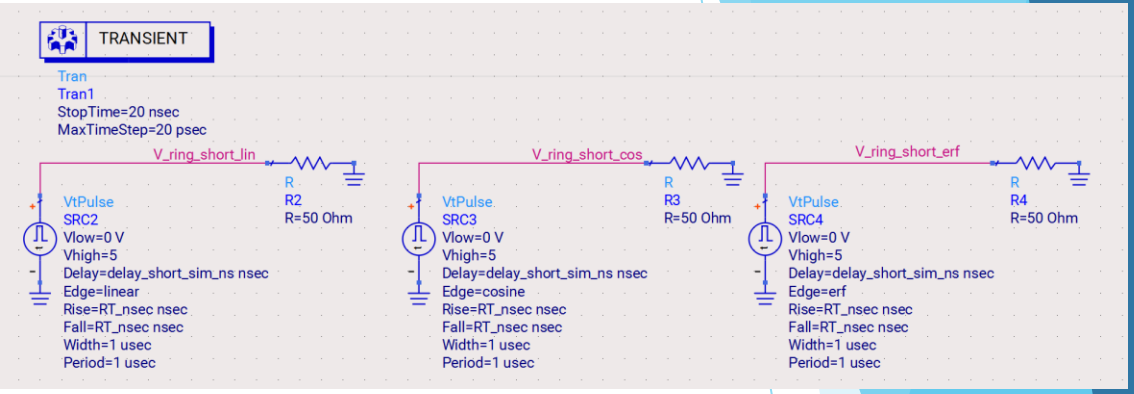
File	Rows	Step	Span	Header
0.5ftRingingHoldoff.txt	2000	0.25 ns	0–499.5 ns	v(1)
3ftRingingHoldoff.txt	2000	0.25 ns	0–499.5 ns	v(1)
0.5ft / 3ft Ringing1.3-2.txt	2000	0.5 ns	0–999.5 ns	v(1)
0.5ftRinging1.3.txt	2000	1 ns	0–1999 ns	v(1)
3ftRinging1.3.txt	2000	1 ns	0–1999 ns	voltage(1)

The ADS Model



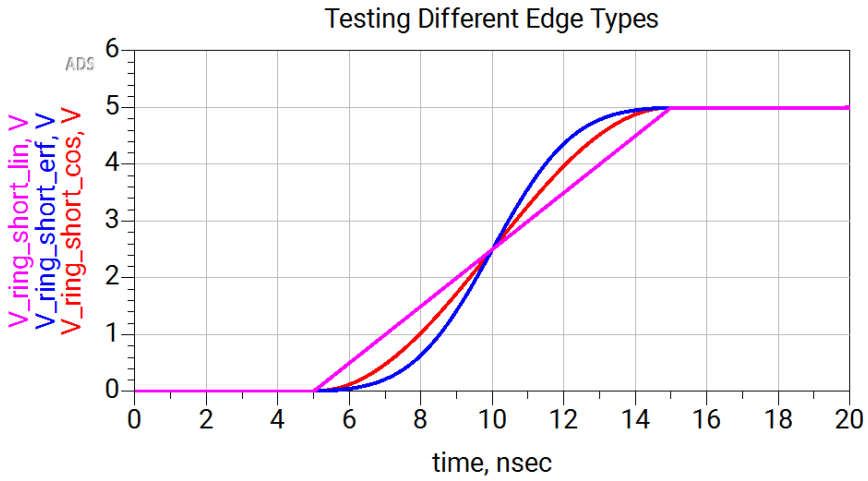
- Model includes a Thevenin resistance, ideal transmission line, and a model of the scope's added impedance
- Nominal values were used based on prior knowledge of the system and measurements taken

Simulated Edge Types



Edge type simulation setup

- Linear, Cosine, and ERF (error function)
  - Different type for different applications
  - ERF replicates a real channel, however is more computationally expensive to simulate



Difference between different edge types
Table to the right illustrates which rising edge is which

Edge type	10% time (ns)	90% time (ns)	Total 10-90 RT	K constant RT
ERF (Blue)	7.766	12.27	4.504	2.220248668
Cosine (Red)	7.066	12.97	5.904	1.693766938
Linear (Pink)	6.006	14.01	8.004	1.249375312

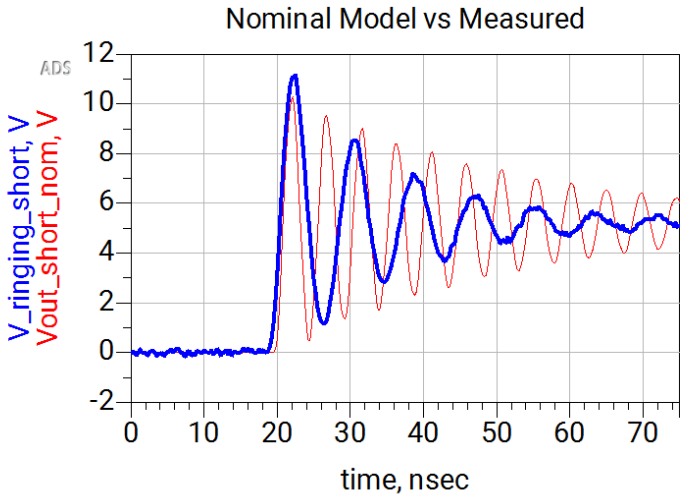
10-90 rise times for different edge selection
The constant K is determined to be about 2.22-2.3 based on these results

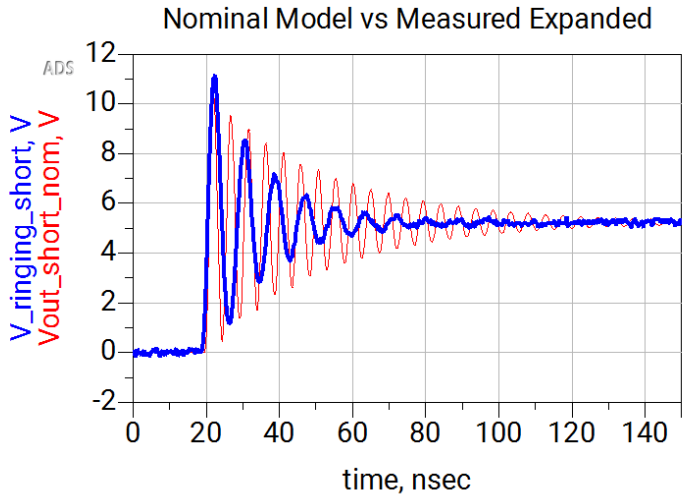
Nominal Model vs Measured

- Nominal values were used based on prior knowledge of the system
  - Thevenin resistance was measured in prior lab
  - Rise time was approximated to be 1 nano second
  - The line delay was calculated based on the data sheet for the coax cable
  - Scope resistance is high, 10 Mohm looks like an open circuit
  - 10X probe has a spec of 9.5 pF

	First order
Rth (ohm)	1.7
Rise time (ns)	1
Line Impedance (ohm)	50
Line Delay (ns)	0.77
Scope Resistance (Mohm)	10
Scope capacitance (pF)	9.5

Nominal values used for the model





Nominal model (red) does not match great with the measured values (blue)

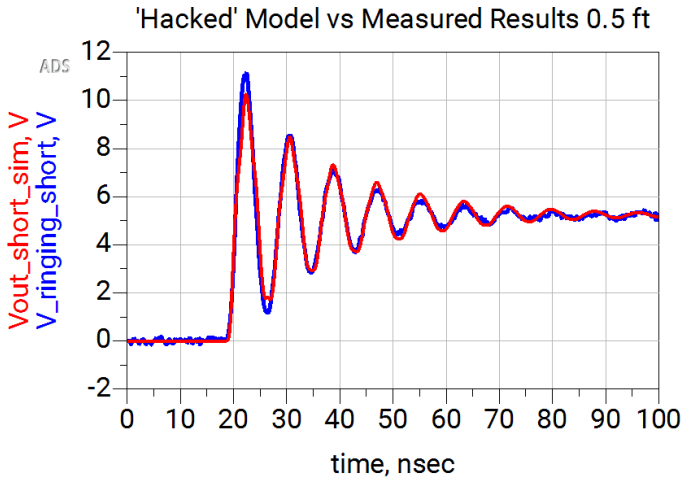
Refined Model vs Measured

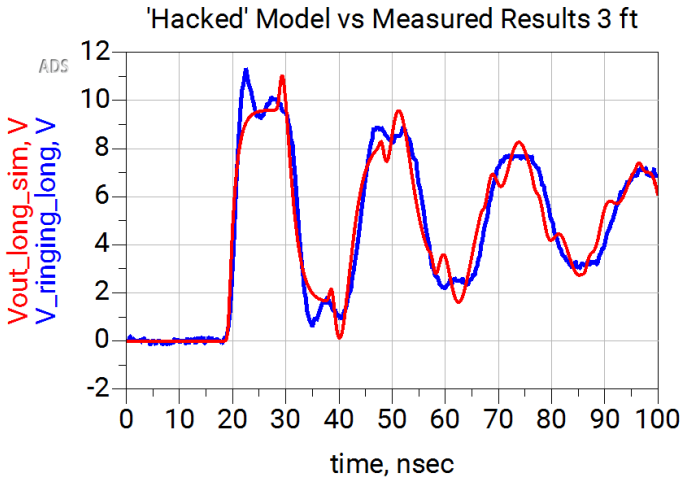
- After further scrutiny of the system, the model could be refined in order to get a closer fit to the measured values
  - Thevenin resistance could be refined based on the measured values, the decay between the periods of ringing implies the resistance (or rather impedance) is higher than expected
  - We know the rise time of the hex inverter is likely faster than what we approximated at first
  - Additional capacitance and delay is introduced by interconnects (connectors, adapters, scope probe lines, etc.)

	First order	Second Order	'Hacked'
Rth (ohm)	1.7	~3-5	4.55
Rise time (ns)	1	>1	0.72
Line Impedance (ohm)	50	47-53	50
Line Delay (ns)	0.77	~1	1.11
Scope Resistance (Mohm)	10	10	10
Scope capacitance (pF)	9.5	>>9.5	22.7

Are the Hacked Values Unique?

- By tuning the parameters, we were able to get a fairly well matched model to the measured system
- The second order approximations of the system led to a decent range at which to tune

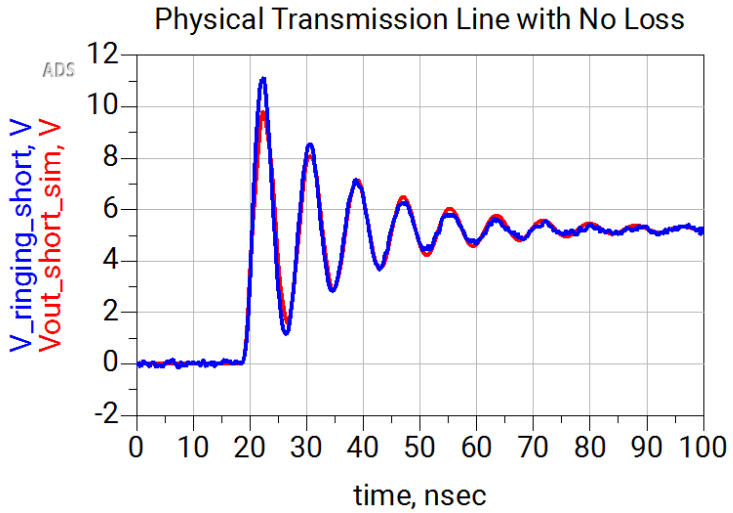


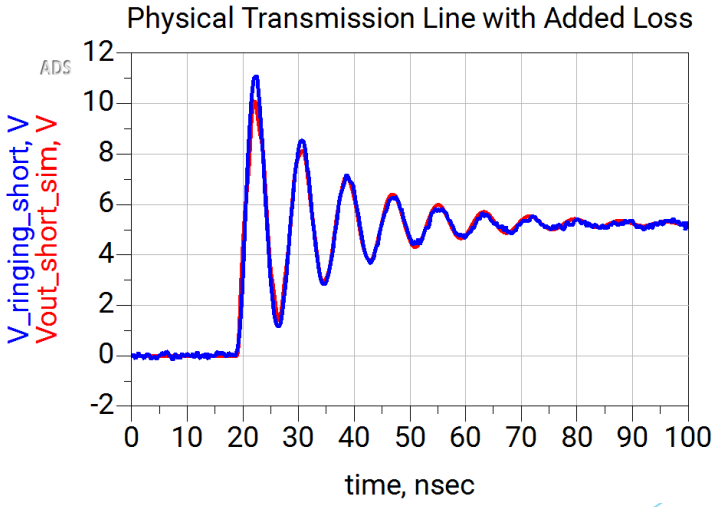


- By tuning the parameters, a good fit was found for the 6 in cable, however the fit for the 3 foot cable did not match as closely
  - However the approximation was still relatively good using the same model not optimized for the individual line

Modelling Non-ideal Cable

- A model with a physical representation of a transmission line can get close to the physically measured system
  - The physical model introduces Dk, attenuation, and loss tangent which all real cables have
- First, the model is best fit without using loss tangent
- Then, the loss was added, a slightly better match was achieved
  - Attenuation had a bigger effect on the decay of the ringing no matter the parameters of the transmission line
  - Loss tangent only had an effect when the rise time was very fast, where frequency was effectively higher





So What?

- A good understanding of the system, including the measurement system, is ideal for optimizing a model in simulation
- The ideal world of software does not match perfectly with real world measurements
  - However, we can get close with a higher order models that includes losses, frequency dependence, and deeper understanding
- Nominal values for a model can get us close to an ideal model but are best used as starting point for refinement of the model
  - The deeper the model, the more generally it can be applied under certain circumstances

Questions?